



TUDOR ANDREI HĂLĂȘAG

I build things. From audio encryption tools and OOP transit simulations to embedded hardware with Arduino — I've been engineering solutions independently since before university. Experienced in team leadership and operations management from running shift operations in a high-volume service environment. Looking to apply that hands-on engineering mindset in a real industrial context.

PROJECTS & PERSONAL DEVELOPMENT

Audio File Encryption/Decryption · *Python* — Built a file-level audio encryption/decryption tool using SHA-256 hashing — full implementation from scratch.

Public Transport Ticketing System · *C++* — Designed an OOP simulation of Bucharest's transit network (metro, tram, bus) with ticket types, passes, and expiry logic.

Chess Constraint Solver · *C++* — Implemented the N-Queens combinatorial problem and related constraint-satisfaction algorithms.

CV Generator Tool · *C#* — Developed a desktop utility for automated document generation with structured data input.

Step-Down Transformer Design · *Engineering* — Designed a functional step-down transformer from scratch, applying electromagnetic theory and component selection.

Arduino & Embedded Systems Lab · *Hardware* — Programmed microcontrollers and interfaced LCD displays, sensors, stepper motors, and speakers — hands-on embedded I/O and memory management.

EDUCATION

BSc - Faculty of Robotics and Industrial Engineering 2022 - Present
Polytechnic University of Bucharest, Bucharest

- Control systems, robotics, and industrial automation — with hands-on lab projects.
- Systems-thinking applied to real engineering problems across hardware and software disciplines.

Mathematics - Computer Science 2017 - 2021
"MIHAIL KOGĂLNICEANU" NATIONAL COLLEGE, Galați

Final Exam: 8.76 / 10

- Intensive track in advanced mathematics and programming.
- Practical C++ and Python from the earliest stages — applied in personal projects throughout.

EMPLOYMENT HISTORY

Barista & Operations Lead Mar 2023 - Sept 2023
5 TO GO EUROPE HOLDING SRL Romania, Bucharest

- Led shift operations: team coordination, opening/closing procedures, and workflow management in a high-volume environment.
- Managed all transactions and inventory via Comax Smart Retail (ERP/POS) and Excel — 100% cash-handling accuracy throughout.
- Drove a 15–20% improvement in peak-hour service efficiency through process restructuring and preparation systems.

Personal

Name
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Skills

C / C++
Python
Microsoft Office Suite
MATLAB
Arduino
AutoCAD Mechanical
LabVIEW
ORACLE SQL
C# / Java
Adobe Illustrator
Tina-TI / LTspice

Languages

Romanian	Native
English	C2
Italian	A2
Turkish	A2

Interests

Music (drums, guitar, bass, DJ)
Sports — gym & outdoor fitness
Travel & hiking trips